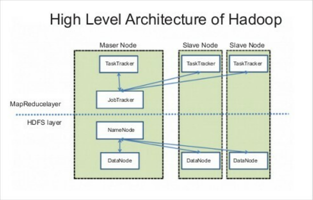

Figure 1: High level architecture

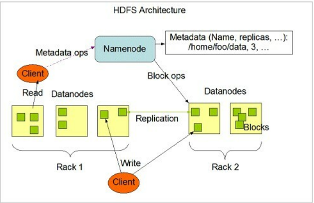

Figure 2: Hadoop architecture

Big Data is a term used to describe large collections of data (also known as data sets) that may be unstructured, and grow so large and so quickly that it is difficult to manage with regular database or statistical tools.

In terms of numbers, what are we looking at? How BIG is Big Data? Well there are more than 3.2 billion Internet users, and active cell phones have crossed the 7.6 billion mark. There are now more in-use cell phones than there are people on the planet (7.4 billion). Twitter processes 7TB of data every day, and 600TB of data is processed by Facebook daily. Interestingly, about 80 per cent of this data is unstructured. With this massive amount of data, businesses need fast, reliable, deeper data insight. Therefore, Big Data solutions based on Hadoop and other analytic software are becoming more and more relevant.

Open source projects related to Hadoop

Here is a list of some other open source projects related to Hadoop:

- *Eclipse* is a popular IDE donated by IBM to the open source community.
- *Lucene* is a text search engine library written in Java.
- *Hbase* is a Hadoop database - *Hive* provides data warehousing tools to extract, transform and load (ETL) data, and query this data stored in Hadoop files.
- *Pig* is a high-level language that generates MapReduce code to analyse large data sets.
- *Spark* is a cluster computing framework.
- *ZooKeeper* is a centralised configuration service and naming registry for large distributed systems.
- *Ambari* manages and monitors Hadoop clusters through an intuitive Web UI.
- *Avro* is a data serialisation system.
- *UIMA* is the architecture used for the analysis of unstructured data.
- *Yarn* is a large scale operating system for Big Data applications.
- *MapReduce* is a software framework for easily writing applications that process vast amounts of data.

Hadoop architecture

Before we examine Hadoop's components and architecture, let's review some of the terms that are used in this discussion. A node is simply a computer. It is typically non-enterprise, commodity hardware that contains data. We can keep adding nodes, such as Node 2, Node 3, and so on. This is called a rack, which is a collection of 30 or 40 nodes that are physically stored close together and are all connected to the same network switch. A Hadoop cluster (or just a 'cluster' from now on) is a collection of racks.

Now, let's examine Hadoop's architecture—it has two major components.

1. *The distributed file system component:* The main example of this is the Hadoop distributed file system (HDFS), though other file systems like IBM Spectrum Scale, are also supported.
2. *The MapReduce component:* This is a framework for performing calculations on the data in the distributed file system.

HDFS runs on top of the existing file systems on each node in a Hadoop cluster. It is designed to tolerate a high component failure rate through the replication of the data. A file on HDFS is split into multiple blocks, and each is replicated within the Hadoop cluster. A block on HDFS is a blob of data within the underlying file system (see Figure 1).

Hadoop distributed file system (HDFS) stores the application data and file system metadata separately on dedicated servers. NameNode and DataNode are the two critical components of the HDFS architecture. Application data is stored on servers referred to as DataNodes, and file system metadata is stored on servers referred to as NameNodes. HDFS replicates the file's contents on multiple DataNodes, based on the replication factor, to ensure the reliability of data. The NameNode and DataNode communicate with each other using TCP based protocols.

The heart of the Hadoop distributed computation platform is the Java-based programming paradigm MapReduce. Map or Reduce is a special type of directed acyclic graph that can be applied to a wide range of business use cases. The Map